Combination circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Reduce a circuit by identifying genuine series and parallel groups.

SUCCESS CRITERIA - I can...

1. Identify nodes before grouping resistors.
2. Reduce one group at a time.
3. Check current and voltage conservation in the original circuit.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “parallel circuit” to explain your model or result. Sentence starters are optional.

series: connected along one unbranched current path

parallel circuit: branches connected across the same two nodes

node: a shared electrical connection point

01 Bring back an earlier idea

Measured (I, V) pairs are (0.10, 3), (0.20, 6), (0.30, 9) in A, V.
Find resistance and explain the evidence for an ohmic model.

Word help: resistance = voltage divided by current for a stated operating condition

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $R_{series} = R_1 + R_2$; $1/R_{parallel} = 1/R_1 + 1/R_2$; $V = IR$. Reduce one known group at a time. Check nodes before calling components series or parallel.

Worked reasoning: A 2 ohm resistor is in series with two 4 ohm resistors in parallel. The parallel pair equals 2 ohms; total 4 ohms. A 12 V supply gives 3 A total.

Your turn: Draw the network and label the two series sections before splitting the branch current.

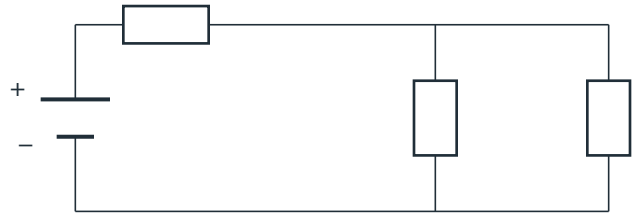
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

A 2 Ω resistor is in series with two parallel 6 Ω resistors.
Draw the arrangement and circle the parallel group.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Solve and justify

For that circuit with a 10 V battery, calculate the parallel equivalent.
Find the series drop, branch voltage and each branch current.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student adds all three resistor values because they are drawn side by side. What should determine the grouping instead?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

06 Choose a method

Two targets each have range 600 m but different radial velocities.
Explain how they can have equal echo delays yet different Doppler shifts.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

Replace the two 6 Ω branches with 3 Ω and 6 Ω. Keep the 2 Ω series resistor and 10 V supply.
Find total current.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 3 ohm series resistor feeds parallel 6 and 3 ohm branches from 15 V.
Find total current, voltage across parallel section and each branch current.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3